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Question	Answer	Marks
1	**Defining the problem**	
	P is the independent variable and θ is the dependent variable, or vary P and measure θ. (Allow θ is the independent variable and P is the dependent variable.)	**1**
	keep density of salt solution constant or keep σ constant	**1**
	Methods of data collection	
	labelled diagram of workable experiment including: • sealed container e.g. bell jar, sealed conical flask • tube connected to pump (for changing pressure) or other workable method • salt solution, labelled	**1**
	workable method to heat salt solution within the sealed container, e.g. hot plate/(electrical) heater	**1**
	description of pressure gauge or manometer to measure P	**1**
	use of a thermometer to measure θ **or** labelled thermometer (in salt solution) in diagram	**1**
	Method of analysis	
	plot a graph of lg θ against lg P (or ln θ against ln P)	**1**
	q = gradient	**1**
	$k = 10^{y\text{-intercept}} / \sigma$ (for ln θ against ln P: $k = e^{y\text{-intercept}} / \sigma$)	**1**

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Question	Answer	Marks
	Additional detail including safety considerations	**Max. 6**
	D1 safety precaution relating to pressure, e.g. safety screen	
	D2 use of (protective) gloves to handle hot salt solution/beaker/flask	
	D3 density of salt solution or ρ given by m / V	
	D4 measuring cylinder to measure volume **and** difference in top pan balance/scales readings to measure mass of salt solution	
	D5 slowly/gradually increase/decrease the temperature/pressure	
	D6 measure P and θ when salt solution starts to boil	
	D7 $\lg\theta = q\lg P + \lg k\sigma$	
	D8 relationship valid if a straight line	
	D9 identification of when (salt solution) starts to boil, e.g. wait until (vapour) bubbles are on the surface or surface moves or temperature remains constant (for heating methods)	
	D10 recheck that density of salt solution is constant/add water to keep density constant/take multiple solutions from a large volume	

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